

Anti-Fraud System - Latency Arbitrage Module

Executive Objective

Transform the existing CRM Anti-Fraud module from a fragmented, predominantly manual analytical system into a holistic, deterministic and event-driven detection, decision-making and control framework, using Latency Arbitrage as the first priority use case. The existing AI/ML behavioural profiling capability will be calibrated in parallel and integrated into the Risk decision workflow.

Current Problem

Current limitation	Impact
Limited event-driven detection	The system was not originally built around configurable event triggers and threshold-based detection.
Limited configurable controls	There is insufficient ability to customise rules, thresholds and escalation criteria within the Anti-Fraud workflow.
No advanced behavioural profiling	Current analysis relies heavily on manual mark-out tables and does not provide a consolidated view of risk, confidence, consistency, persistence or economic impact.
Fragmented functionality	Modules and analytical outputs are scattered, making the overall risk picture difficult to assess in one place.
Manual investigation	Risk users must manually identify, reconstruct, compare and interpret many cases.
No consolidated decision layer	Individual indicators are not consistently combined into a structured Risk assessment.
Limited automation pathway	Detection is not naturally connected to controlled downstream actions.

Three-Week Plan - ETA 27 September 2026

Period	
Week 1 — Current State, Requirements & Technical Architecture 7–13 September	Map the existing Anti-Fraud modules and workflow detection requirements, event triggers, configurable and market/reference-data timestamps and historic high-volume data interaction. Review the existing A Trading DevOps to assess the architecture, process CRM Anti-Fraud module and the Trade Server / Execution Fraud alerts will be consumed by the execution engine
Week 2 — Build Detection, Decisioning & Automation 14–20 September	Develop the event-driven Latency Arbitrage detection logic, mark-outs, client aggregation and behavioural scoring. Integrate the AI/ML behavioural outputs into action layer, allowing validated alerts to be passed Smart Order Routing (SOR) actions, with defined routing/execution treatment. Implement audit logging
Week 3 — Validation, Calibration & Production Deployment Readiness 21–27 September	Conduct historical replay and validate known/misclassified controlled live/shadow testing. Validate end-to-end Engine workflow. Tune thresholds and scoring. Validate confirmed cases. Test system performance under execution and manual override controls. Complete Risk and T

Expected Outcome

At the end of three weeks, the organisation should have a validated first version of a modern Anti-Fraud workflow, with Latency Arbitrage as the priority capability. The deliverable is therefore broader than a detection model: it is the foundation for moving the CRM Anti-Fraud module from fragmented manual analysis toward integrated detection, decision-making and controlled automation for risk management actions.

Measurables Target

Headline KPI	27 Sep Initial Target
Detection Rate	≥80%
Detection Improvement vs. Baseline	≥30%
Alert Accuracy	≥70%
Automation Success – SOR / Execution	≥95%
Manual Intervention Reduction	≥50%

Task Allocation and Tracker

 Different views are available

 All

☐  Task	 Task owner
1 Technical requirements for Latency Arbitrage	☐ Reginald Sherekete ☐ Reza Sadeghi ☐ Jason
2 Review Technical Architecture	☐ Igor Merkulyev
3 Build the Detection and Profiling Engine	☐ Roy Vivasi
4 Integrate the Alerts to Antifraud System	☐ Igor Merkulyev
5 Build the Action Layer for Smart Order Routing	☐ Igor Merkulyev
6 Validation and Calibration	☐ Reza Sadeghi ☐ Teoh Xue Dong ☐ Jason
7 Data Sources and Database Architecture	☐ Charles Lin ☐ Will Li ☐ Demsy Iman Must

7 records

Required Resources

Resource	Required Allocation	Responsibility
Anti-Fraud Team	3 members	Anti-Fraud requirements, detection and operational acceptance
Quant Team	2 members	Detection methodology, statistical analysis and validation of detection accuracy
Data Team	2 members	Data availability, data preparation, data requirements
Development / Engineering	To be advised	Development, system integration, integration and SOR/Execution Engine
Trading DevOps	Dedicated support as required	Real-time processing, infrastructure deployment and production readiness
Other Departments	As required	Each relevant department to provide expertise necessary to meet the 27 September 2020 deadline